JECRC UNIVERSITY JAIPUR

Session -2025-26

Subject: Mathematics-I (DMA024A)

Course: B. Tech – I Semester

Tutorial Sheet – 14 (CO5)

UNIT-5

CO5: To acquaint the student with mathematical tools needed in evaluating multiple integrals and their usage.

PART – A **(5X2=10)**

A1: (CO5) Define centre of gravity.

A2: (CO5) Define volume integral.

A2: (CO5) Evaluate $\int_{-1}^{1}\int_{-2}^{2}\int_{-3}^{3} dxdydz$. **Ans:** 48

A4: (CO5) Evaluate $\int_{0}^{4}\int_{0}^{x}\int_{0}^{x+y} zdzdydx$. **Ans:** 70

A5: (CO5) Evaluate $\int_{0}^{1}\int_{0}^{1}\int_{0}^{1}(x^2+y^2+z^2)dzdydx$. **Ans:** 1

PART – B **(3X7=21)**

B1: (CO5) Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum.

Ans: $8a^2/3$

B2: (CO5) Find the mass of the tetrahedron bounded by the co-ordinates planes and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c} = 1$, the variable density $4xyz$. **Ans:** $a^2b^2c^2/180$

B3: (CO5) Find centre of gravity of the area in the positive quadrant of the curve $x^{2/3} + y^{2/3} = 2^{2/3}$. **Ans:** $(512/315\pi, 512/315\pi\ )$

PART – C **(3X11=33)**

C1: (CO5) Find the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$. **Ans:** $16a^2/3$

C2: (CO5) Evaluate $\int\int\int \frac{dxdydz}{(x+y+z+1)^3}$ if the region of integration is bonded by the coordinate planes and the plane $x + y + z = 1$. **Ans:** $\frac{1}{2}log2 - \frac{5}{16}$

C2: (CO5) Evaluate $\int\int\int x^2\, dxdydz$ over volume of tetrahedron bounded by $x = y = z = 0$ and $\frac{x}{a}+\frac{y}{b}+\frac{z}{c} = 1$. **Ans:** $\frac{a^3bc}{60}$